

MASS ADAS L3 — Verification & Validation Plan v0.1

2026-05-21

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MASS ADAS L3 Tactical Decision Layer — Verification & Validation Plan v0.1

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Revision History

Version	Date	Author	Description
v0.1	2026-05-31	V&V Engineer	Initial release. Phase 1–4 gate structure, KPI matrix, coverage dimensions, SIL latency budget, RL rebound path, DNV toolchain entry conditions, certification evidence framework, DEMO-1 scope.

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1. Purpose & Scope

This Verification & Validation Plan defines the strategy, criteria, and infrastructure for demonstrating that the MASS ADAS L3 Tactical Decision Layer meets its safety, functional, and performance requirements across the full development lifecycle from first integration through sea trial.

Scope boundaries:

- **In scope:** All eight TDL modules (M1 ODD/Envelope Manager through M8 HMI/Transparency Bridge), the SIL/HIL test infrastructure, the scenario generation and coverage tracking framework, the certification evidence pipeline, and the end-to-end performance KPIs.
- **Out of scope:** L1 Mission Layer, L2 Voyage Planner, L4 Guidance Layer, L5 Control & Allocation, Multimodal Fusion, Cybersecurity hardening, and shore-side infrastructure. These layers are treated as test fixtures or boundary conditions.

This plan covers the period from Phase 1 (DEMO-1 Skeleton Live, 15 June 2026) through Phase 4 (HIL integration and third-party SIL 2 assessment, September–November 2026). It is structured in accordance with DNV-CG-0264 Section 3 (Verification Plan) conventions.

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2. V&V Strategy Overview

2.1 Three-Phase Verification Approach

Verification proceeds through three strictly sequential phases. Each phase has formal entry and exit gates defined in Section 3. No phase may begin until the preceding phase's exit gate has been cleared.

Phase	Primary Tool	Time Scale	Focus	Key Deliverable
1. SIL	Software-in-the-loop simulation	0.02–1 s timestep	Functional correctness, COLREGs compliance, coverage, performance KPIs	SIL exit report with scenario pass rates, coverage cube, KPI validation
2. HIL	Hardware-in-the-loop with real-time target	Real-time (at least 10 Hz)	Real-time determinism, DDS latency, hardware driver integration, M7 watchdog on target	HIL exit report with jitter measurements, determinism evidence
3. Sea Trial	FCB non-certification sea trial	Full mission duration	End-to-end behavior in real environmental conditions, ROC HMI validation, ASDR logging integrity	Sea trial report with encounter logs, KPI replay, anomaly registry

2.2 SIL as Primary Verification Tool

SIL remains the primary verification vehicle for Phases 1 through 3. The majority of functional and safety verification is performed in SIL because scenario space is enumerable and reproducible, fault injection is deterministic, coverage metrics can be collected automatically, and

COLREGs compliance can be evaluated against geometric benchmarks with ground-truth geometry.

HIL and sea trial serve as confirmation activities. They do not discover new functional defects; they verify that the real-time implementation, hardware integration, and environmental noise do not degrade the behaviors already proven in SIL.

2.3 Independence of V&V

The V&V function is organizationally independent from the development team:

- The V&V Engineer (FTE, 5/8–8/31) owns this plan, the scenario schema, the coverage methodology, and the gate criteria.
- V&V writes test scenarios, operates the SIL framework, and adjudicates gate pass or fail.
- Developers do not modify gate criteria, coverage thresholds, or scenario definitions without V&V review.
- Safety Engineer (outsourced, 5/15–7/10) reviews V&V results for SOTIF and IEC 61508 alignment but does not execute tests.

3. Entry/Exit Gates

3.1 Phase 1 — SIL Entry Gate (E1)

The SIL entry gate ensures that the development baseline is stable enough to yield meaningful verification results.

ID	Criterion	Threshold	Evidence Artifact
E1.1	All D1.x tasks closed per v3.0 DoD	100% closed	D1.x task tracker export
E1.2	colcon build clean on CI	0 errors, 0 warnings treated as errors	CI build log
E1.3	CI pipeline green	All jobs pass	GitLab pipeline status
E1.4	Scenario schema validated	Schema v1.0 passes structural validation	schema_validate.py report
E1.5	Mock publisher frequencies verified	All cross-layer mock pubs within plus or minus 5% of specified rate	Frequency check log
E1.6	E2E data flow sanity	M1 to M2 to M4 to M5 to M8 message chain completes in under 5 s	E2E latency trace
E1.7	V&V Plan v0.1 committed	File present in docs/Design/V&V_Plan/	Git commit hash
E1.8	M7 watchdog Python stub at least 1 PASS	At least 1 pytest test with m7 in nodeid passes	pytest-report.json

Gate E1.1 through E1.8 are checked by `tools/check_entry_gate.py --phase 1`. Manual checks (e.g., design review sign-off) are recorded in the Phase 1 gate log but do not block automated execution.

3.2 Phase 1 — SIL Exit Gate (X1)

ID	Criterion	Threshold	Evidence Artifact
X1.1	Baseline scenario pass rate	At least 95% of 50 baseline scenarios pass	test-results/baseline_results.json
X1.2	KPI matrix validated	All Section 4 KPIs within threshold on at least 30 scenario runs	test-results/kpi_matrix.json
X1.3	ASDR log consistency	Every decision cycle logged with timestamp, module state, rationale	ASDR audit script report
X1.4	Coverage cube minimum	At least 10 of 1100 cells lit (Phase 1 baseline)	test-results/coverage_cube.json
X1.5	Imazu-22 benchmark	22 of 22 pass, CPA at least 200 m ratio at least 95%, COLREG classification at least 95%	test-results/imazu22_results.json
X1.6	M7 watchdog critical-path coverage	All 7 monitored modules exercised for timeout and recovery	pytest-report.json

3.3 Phase 2 — HIL Entry Gate (E2)

ID	Criterion	Threshold	Evidence Artifact
E2.1	SIL exit gate cleared	X1.1 through X1.6 all pass	Phase 1 exit report
E2.2	HIL hardware qualified	Target hardware flashed and booting	HIL qualification checklist
E2.3	Real-time constraints verified	ROS2 timer jitter P95 under 5 ms on target	ros2 topic hz plus trace analysis
E2.4	DDS-Security profile loaded	Authentication and encryption active on test domain	dds_security.xml validation

3.4 Phase 2 — HIL Exit Gate (X2)

ID	Criterion	Threshold	Evidence Artifact
X2.1	Deterministic replay	Same scenario input produces same output within 1% on HIL versus SIL	Replay diff report
X2.2	Latency budget held	dds-fmu bridge P95 under 10 ms, P99 under 15 ms	Latency histogram CSV
X2.3	M7 watchdog on target	Watchdog triggers MRC within 3 timeout periods on hardware	Target test log
X2.4	200 scenarios run	At least 200 scenarios executed on HIL with at least 95% pass	hil_results.json

3.5 Phase 3 — Sea Trial Entry Gate (E3)

ID	Criterion	Threshold	Evidence Artifact
E3.1	HIL exit gate cleared	X2.1 through X2.4 all pass	Phase 2 exit report
E3.2	CCS preliminary approval	CCS surveyor sign-off on ConOps and V&V Plan	CCS letter
E3.3	Crew training complete	ROC and master completed simulator familiarization	Training completion certificates
E3.4	Emergency stop chain tested	Hardware override plus ASDR zero-software e-stop verified on dock	Dockside test log

3.6 Phase 3 — Sea Trial Exit Gate (X3)

ID	Criterion	Threshold	Evidence Artifact
X3.1	Mission duration	At least 50 nm and at least 100 h autonomous operation	VDR log plus AIS track
X3.2	Encounter events	At least 10 live COLREGs encounters with target vessels	ASDR encounter registry
X3.3	No safety incidents	Zero MRC activations attributable to TDL defect	Incident registry (empty)
X3.4	KPI replay	All Section 4 KPIs within 2 times threshold when replayed from sea trial logs	Post-processed replay report

4. End-to-End KPI Matrix

The following KPIs are measured end-to-end from the first L2 waypoint input to the L4 command output. Measurements are taken under nominal conditions (open sea, Beaufort 0–1, full sensor availability) unless otherwise noted.

KPI	Target	Measurement Method	Phase Verified
AvoidancePlan P95 latency	Less than or equal to 1.0 s	colcon test timing wrapper on M4 to M5 to M8 chain	SIL Phase 1
ReactiveOverrideCmd P95	Less than or equal to 200 ms	colcon test timing wrapper on M7 override path	SIL Phase 1
Mid-MPC solve time	Under 500 ms	M5 solver diagnostics (OSQP / ECOS callback)	SIL Phase 2
BC-MPC solve time	Under 150 ms	M5 solver diagnostics (BC horizon)	SIL Phase 2
M7 safety check	Under 10 ms	M7 standalone benchmark (Python stub to C++ target)	SIL Phase 1
M4 arbitration cycle	Under 100 ms	M4 lvP diagnostics (objective function evaluation plus winner selection)	SIL Phase 2

Measurement protocol:

1. Each KPI is sampled over 1000 consecutive decision cycles.
2. P95 and P99 are computed from the empirical CDF.
3. Outliers beyond P99 are logged as anomalies and reviewed by V&V.
4. If any KPI exceeds its target, the responsible module's D-task is reopened for performance regression analysis.

5. Test Coverage Dimensions

5.1 Functional Coverage

Dimension	Granularity	Phase 1 Target	Phase 2 Target	Phase 3 Target
Module core scenarios	8 modules times 5 core scenarios each	40 scenarios	40 scenarios	40 scenarios
ODD zone variants	4 zones (open sea, coastal TSS, port approach, offshore wind farm)	2 zones	4 zones	4 zones
Functional total	8 times 5 times zones	80 runs	160 runs	160 runs

5.2 Performance Coverage

- Latency KPIs (Section 4) measured under varying load: 1, 5, 10, 20 concurrent targets.
- Memory footprint: RSS monitored every 10 s during 1 h continuous run.
- CPU utilization: per-core load on target hardware (HIL) or host (SIL).

5.3 Failure Response Coverage

M7 Safety Supervisor enforces 6 hard constraints. Each constraint is tested against 10 failure scenarios:

Hard Constraint	Failure Scenarios (examples)
HC-1: CPA at least min safe distance	Sensor dropout, target ghost, speed estimate error
HC-2: No ODD violation	ENC mismatch, GPS spoofing, weather threshold breach
HC-3: MRC trigger within TMR	Comms loss to shore, M1 deadlock, DDS partition fault
HC-4: COLREGs rule never broken	Rule ambiguity, multi-target conflict, crossing plus overtaking
HC-5: Speed limit respected	Engine fault, current overspeed, waypoint overshoot
HC-6: M8 transparency never silent	M8 crash, logger disk full, DDS topic overflow

Total failure response test cases: 6 times 10 = 60 cases.

5.4 COLREGs Compliance Coverage

The COLREGs compliance matrix is defined as:

- 11 rules (Rule 5, 6, 7, 8, 9, 13, 14, 15, 16, 17, 19)
- 4 ODD zones
- 5 disturbance levels (Beaufort 0–1, 2–3, 4–5, 6–7, sensor degraded)
- 5 PRNG seeds per cell

Total cells: 11 times 4 times 5 times 5 = **1100 cells**.

Phase 1 lights at least 10 cells (baseline). Phase 2 lights at least 200 cells.

Phase 3 lights at least 1000 cells (full matrix minus contingency cells reserved for edge cases).

5.5 Scenario Distribution Summary

Phase	Scenario Count	Source	Coverage Focus
Phase 1 (DEMO-1)	50	Imazu-22 plus synthetic baseline	Infrastructure validation, schema maturity
Phase 2 (DEMO-2)	200	Expanded Imazu variants plus COLREGs matrix	Decision quality, M4/M5 maturity
Phase 3 (DEMO-3)	1000+	Full 1100-cell matrix plus 100+ stress cases	SIL 2 evidence, SOTIF completeness
Phase 4 (HIL)	200 replayed from SIL	Deterministic subset of Phase 3	Real-time confirmation
Phase 4 (Sea Trial)	10+ live encounters	Unscripted maritime traffic	Environmental validation

6. SIL Latency Budget

The SIL toolchain introduces synthetic latency that must be bounded and compensated so that SIL results remain valid predictors of HIL and sea trial behavior.

6.1 dds-fmu Bridge Latency

Metric	Budget	Measurement Point
P95 latency	Under or equal to 10 ms	ROS2 topic timestamp to FMU input port ready
P99 latency	Under or equal to 15 ms	Same as above
Max jitter	Under or equal to 5 ms	Absolute deviation from mean per 1 s window

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If the dds-fmu bridge exceeds P99 15 ms, the V&V Engineer shall:

1. Increase the DDS QoS reliability buffer depth.
2. Enable jitter buffering in the FMI co-simulation master.

3. If steps 1 and 2 do not restore the budget, escalate to the Safety Engineer for a latency budget revision.

6.2 M7 Native ROS2 Boundary

M7 Safety Supervisor is strictly ROS2 native. It does **not** cross the FMI boundary. This eliminates dds-fmu latency from the safety-critical path and guarantees that M7's VETO decision is always based on the freshest possible world model.

6.3 Jitter Buffer Compensation

The SIL framework implements a jitter buffer on each FMU input port:

- Buffer depth: 3 samples (configurable per scenario).
- Timeout: 2 times mean measured jitter; if exceeded, the simulation step is marked DEGRADED and logged.
- Scenarios with more than 1% DEGRADED steps are excluded from pass-rate statistics until the latency root cause is resolved.

7. RL Artefact Rebound Path

Phase 4 includes reinforcement learning adversarial testing (D4.6, October–December 2026). To prevent a toolchain redesign in Phase 4, the RL artefact rebound path is specified now and locked for the project duration.

7.1 Pipeline

```
RL training (Stable Baselines3 / Ray RLlib)
  ↓ export
ONNX model (policy network)
  ↓ convert
mlfmu wrapper (ONNX Runtime inside FMU 2.0)
  ↓ package
FMU 2.0 co-simulation unit
  ↓ reinject
libcosim SIL/HIL co-simulation master
  ↓ observe
TDL module under test (M4 or M5)
```

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7.2 Interface Contract

- The ONNX model exposes exactly one input tensor (observation vector) and one output tensor (action vector).
- The mlfmu wrapper maps the action vector to a ROS2 message via the existing l3_msgs IDL (v1.1.2 Section 15.1).
- The FMU step size is 100 ms (10 Hz), matching the M4 arbitration cycle.
- The RL policy does not train during SIL/HIL execution. It is a frozen inference graph.

7.3 Version Lock

Component	Locked Version	License	Rationale
ONNX	1.16+	Apache-2.0	Standard export format from PyTorch and TensorFlow
ONNX Runtime	1.18+	MIT	Jetson-compatible inference engine
mlfm	0.3.x	BSD-3-Clause	DNV-validated ONNX-to-FMU bridge
libcosim	0.10.x	MPL-2.0	OSP co-simulation master

No substitution of these components is permitted without V&V review and a regression test of the full rebound pipeline.

8. DNV Toolchain Entry Conditions

The SIL/HIL toolchain is subject to DNV-RP-0513 (Software in Certification) and DNV-RP-0671 (Model-Based Verification). The following entry conditions apply before the toolchain is accepted for certification evidence generation.

8.1 Version Locking

Tool / Library	Minimum Version	License	Verification Action
maritime-schema	0.2.x	Apache-2.0	Schema validation against v1.0 scenario corpus
libcosim	0.10.x	MPL-2.0	Deterministic replay test (same input produces same output)
farn	0.9.x	MIT	Batch scenario runner regression test
ROS2 Jazzy	2024.04	Apache-2.0	Colcon build plus test on Ubuntu 22.04
FMI standard	2.0	—	FMU compliance checker (FMIChecker)

8.2 License Verification

All third-party tools used in the certification evidence pipeline carry permissive open-source licenses (Apache-2.0, MIT, MPL-2.0, BSD-3-Clause). No GPL or proprietary closed-source tools are used in the evidence generation path. License compliance is verified by:

- `tools/ci/check-licenses.py` scans `pyproject.toml`, `package.xml`, and `CMakeLists.txt` for non-permissive licenses.
- V&V Engineer reviews the scan report before each phase gate.
- Any new dependency added after Phase 1 must pass the license scan before it is used in CI.

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8.3 Self-Qualification Evidence (RP-0513 Section 4)

DNV-RP-0513 Section 4 requires the project to collect evidence that the toolchain itself is fit for purpose. The following evidence is maintained:

Evidence Item	Collector	Storage Location	Review Frequency
Tool version manifest	CI pipeline	test-results/tool_manifest.json	Every build
Deterministic replay log	V&V Engineer	test-results/replay_log/	Every SIL batch
FMU compliance report	fmucheck	test-results/fmu_compliance/	Every FMU release
Schema validation report	schema_validate.py	test-results/schema_validation/	Every scenario commit
License scan report	check-licenses.py	test-results/license_scan/	Every dependency change

9. Certification Evidence Tracking

9.1 Evidence Categories

The CCS i-Ship (Nx, Ri/Ai) application requires evidence across 9 categories. Each category is mapped to D-tasks in the v3.0 development plan.

CCS Category	Description	Mapped D-Tasks	Primary Artifact
C1 — Functional Design	Architecture, interfaces, ODD	D0, D1.1–D1.4, D2.1–D2.4	Architecture report v1.1.2, IDL specs
C2 — Software Lifecycle	SDLC, CM, version control	D1.5, D1.8, D3.9	Git history, CI logs, change records
C3 — Verification Plan	V&V strategy, gates, KPIs	D1.5 (this document)	V&V Plan v0.1–v1.0
C4 — SIL Evidence	Scenario pass rates, coverage, KPIs	D1.6, D1.7, D2.5, D3.6	test-results/JSON suite
C5 — HIL Evidence	Real-time determinism, hardware tests	D4.1, D4.2	HIL exit report
C6 — Safety Analysis	HARA, FMEDA, SOTIF	D2.7, D3.3a, D3.3b	HARA v0.1, FMEDA tables
C7 — Human Factors	ToR, HMI, ROC training	D2.6, D3.5', D4.5'	HF reports, Figma, training records
C8 — Sea Trial	Non-certification trial evidence	D4.5	Trial report, AIS/VDR logs
C9 — Cybersecurity	IACS UR E26/E27, DDS-Security	D3.9	RFC-007, security audit

9.2 Traceability

Every test scenario carries a `cert_evidence_tags` field (array of strings) that links the scenario to one or more CCS categories. The batch runner aggregates these tags into the coverage cube and the final exit report. This ensures that every passing scenario is automatically traced to the certification evidence framework without manual bookkeeping.

9.3 Evidence Maturity

Phase	Evidence Maturity	Action if Gaps Found
Phase 1	C1 through C3 complete, C4 stub (50 scenarios)	Reopen D1.6/D1.7 if coverage under 10% of target
Phase 2	C4 expanded (200 scenarios), C6 stub (HARA v0.1)	Escalate to Safety Engineer if HARA blocks FMEDA
Phase 3	C4 complete (1000+ scenarios), C6 complete (FMEDA), C7 complete	Independent DNV review if pass rate under 95%
Phase 4	C5, C8, C9 complete	CCS preliminary review before sea trial

10. DEMO-1 V&V Scope

DEMO-1 (15 June 2026) is an internal skeleton-live demonstration. Its V&V scope is strictly limited to **infrastructure verification**. Decision-quality verification is explicitly out of scope for DEMO-1 and is the focus of DEMO-2 (31 July 2026).

10.1 In Scope for DEMO-1

Item	Verification Activity	Success Criterion
Build system	colcon build on CI and local dev	Zero errors, under 10 min wall time
CI pipeline	All stage-1 and stage-2 jobs green	100% pass rate
Scenario schema	Schema v1.0 validates 100% of committed scenarios	schema_validate.py exits 0
Mock publishers	All cross-layer mock nodes publish at specified frequency	Plus or minus 5% rate tolerance
E2E data flow	M1 to M2 to M4 to M5 to M8 message chain completes end-to-end	Message received at M8 within 5 s of M1 trigger
Gate automation	check_entry_gate.py – phase 1 exits 0	All automated gates pass
Watchdog stub	M7 Python watchdog passes white-box tests	At least 1 test with m7 in nodeid passes
Imazu-22 baseline	22 benchmark scenarios run and aggregate	CPA at least 200 m ratio at least 95%, COLREG classification at least 95%

10.2 Out of Scope for DEMO-1

The following are verified in later phases and are **not** required for DEMO-1 gate clearance:

- M3 Mission Manager route optimization quality.
- M4 Behavior Arbiter IvP multi-objective trade-off correctness.
- M5 Tactical Planner MPC trajectory optimality.
- M6 COLREGs Reasoner rule interpretation accuracy beyond geometric classification.
- M7 Safety Supervisor hard-constraint enforcement under fault injection.
- M8 HMI/Transparency Bridge SAT-1/2/3 compliance.
- Real-time determinism (HIL scope).

- Environmental robustness (sea trial scope).

10.3 DEMO-1 Gate Checklist

- ☐ E1.1 D1.x tasks 100% closed
- ☐ E1.2 colcon build clean
- ☐ E1.3 CI pipeline green
- ☐ E1.4 Scenario schema validated
- ☐ E1.5 Mock publisher frequencies verified
- ☐ E1.6 E2E data flow under 5 s
- ☐ E1.7 V&V Plan v0.1 committed
- ☐ E1.8 M7 watchdog at least 1 PASS
- ☐ X1.1 50 baseline scenarios at least 95% pass
- ☐ X1.4 Coverage cube at least 10 cells
- ☐ X1.5 Imazu-22 22/22, CPA at least 95%, COLREG at least 95%

Run the automated checks:

```
python tools/check_entry_gate.py --phase 1 --artifacts-dir  
test-results/
```

If the script prints `Phase 1 gate CLEARED`, DEMO-1 V&V scope is satisfied.

Document Approval

Role	Name	Signature	Date
V&V Engineer			
Safety Engineer			
Project Manager			
DNV Validation Lead			